Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100203 | *Subject: Chemistry* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

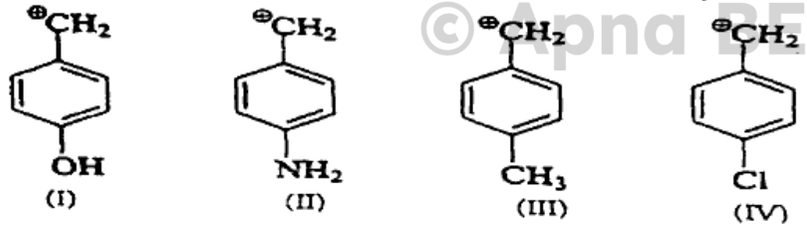
Q.1 Choose the correct answer the following (Any seven question only): [2 x 7 = 14]

(a) How many electrons can exist with the principal quantum number's value as 4?
(i) 16 (iii) 32
(ii) 4 (iv) 12

(b) A ball of mass 0.5kg is moving with velocity 6.626 m/s. What's the wavelength of that ball?
(i) 1×10^{-34} m (iii) 2×10^{-34} m
(ii) 2×10^{-33} m (iv) 2×10^{-3} m

(c) As per Heisenberg's principle of uncertainty, the relation between relative momentum and relative position is _____
(i) Independent (iii) Equal
(ii) Directly proportional (iv) Inversely Proportional

(d) Arrange the following intermediate into decreasing order of stability.



(i) I > II > III > IV (iii) II > I > IV > III
(ii) II > IV > III > I (iv) II > I > III > IV

(e) The distance between the centers of the peaks of doublet is called as?
(i) Coupling constant (iii) Spin-spin coupling
(ii) Spin constant (iv) Chemical shift

(f) How many peaks are expected in low-resolution NMR spectrum of vinyl chloride and ethyl cyclopropane?
(i) 3,5 (iii) 6,3
(ii) 5,3 (iv) 3,6

(g) Which of the following phase changes are endothermic?
(i) Melting and condensation (iii) Vaporization and condensation
(ii) Sublimation and condensation (iv) Melting, sublimation and vaporization

(h) The Gibbs free energy is positive when a change in enthalpy and change in entropy are positive at _____
(i) high temperatures (iii) At All temperatures
(ii) Low temperatures (iv) Only at 0 Kelvin

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(i) The type of hardness that occurs due to the presence of bicarbonate of calcium and magnesium ions hardness is _____.

(i) half hardness
(ii) Temporary hardness
(iii) Permanent hardness
(iv) Momentary hardness

(j) Which of the following is an alkane which can exhibit optical activity?

(i) Neopentane
(ii) Isopentane
(iii) 3–Methyl pentane
(iv) 3–Methyl hexane

Q.2 (a) Write short notes on: [7]
(i) Concept of Atomic Orbitals
(ii) Shapes of s, p and d orbitals

(b) What are Quantum numbers and what do they signify? How are these numbers related with each other? [7]

Q.3 (a) Give reason for the following:- [5]
(i) O_2 is paramagnetic (ii) He_2 does not exists

(b) Find out the bond orders of CO, N_2, O_2^{2-}, O_2^{2+} [5]

(c) Which of the following two molecules has a higher bond length (i) O_2; (ii) O_2^{2+} ? Explain using Molecular orbital Theory. [4]

Q.4 (a) Describe and compare the splitting of d- orbitals under the influence of octahedral and tetrahedral ligand fields. [7]

(b) Explain Crystal Field Stabilization Energy (CFSE) and pairing Energy (PE). How do the CFSE values related to the corresponding colours of d-block metal complexes? [7]

Q.5 (a) Why is methanol a good solvent for U.V. Visible but not for IR spectroscopy? [5]

(b) Calculate the number of vibrational degrees of freedom in the following compounds [5]
(i) CO2, (ii) SO2 , (iii) CH4

(c) What is the importance of fingerprint region in IR spectra? [4]

Q.6 (a) Discuss spin-spin relaxation in NMR spectroscopy and draw and explain the splitting pattern observed in the 1H NMR of $CHCl_2CH_2Cl$. [9]

(b) The frequency of oscillation of HF is 4.12×10^{14} Hz Calculate the vibrational energy and its force constant.M_H =1.0078u M_F =18.9984u. [5]

Q.7 (a) What are Ion exchange Resins? Discuss their application in water softening. How are spent resins regenerated? [7]

(b) 9.6 ml of EDTA solution consumes exactly 10.2 mg of $CaCO_3$ equivalents. 100 ml of a given Hard water sample consumes 17.5 ml of above EDTA solution. 250 ml of the above hard water sample was boiled completely and boiled off sample consumed 35 ml of above EDTA solution. Calculate the temporary and permanent hardness of the above hardwater sample in ^0C, ^0F and as $CaCO_3$ equivalents. [7]

Q.8 (a) Differentiate between terms Configuration and Conformation with suitable examples. [5]

(b) Discuss stereochemistry of tartaric acid. [5]

(c) Explain racemisation of optical isomers with suitable examples. [4]

Q.9 (a) What do you mean by cyclization and ring opening reactions give two examples of each type of these reactions. [7]

(b) What are carbonium ions? Show hybridisation in the structure of carbonium ion and discuss the stability of primary, secondary and tertiary carbonium ions. [7]

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